

MINI PROJECT | TRADING STRATEGY CASE STUDY

Basics of Technical Indicators in Stock Trading

RSI • MACD • Moving Averages

A Comprehensive Case Study with Applied Charts & Practical Trading Rules

CodeAlpha Internship | Stock Market & Trading Domain

Task 4: *Mini Project — Trading Strategy Case Study*

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1. Introduction to Technical Analysis

Technical analysis is the study of past market data — primarily price and volume — to forecast future price movements. Unlike fundamental analysis, which examines a company's financial health, technical analysis operates on the premise that all publicly known information is already reflected in the stock's price, and that price movements follow identifiable patterns and trends.

Technical indicators are mathematical calculations applied to price or volume data. They fall into two broad categories: **trend-following indicators** (like Moving Averages and MACD), which work best in trending markets, and **momentum oscillators** (like RSI), which are more effective in range-bound markets. Skilled traders typically combine both types to reduce false signals and improve decision quality.

Why Technical Indicators Matter

- Provide objective, rule-based entry and exit signals, reducing emotional decision-making
- Help identify the direction, strength, and momentum of a trend
- Can be backtested on historical data to evaluate their effectiveness
- Widely used by retail and institutional traders on NSE, BSE, and global exchanges

2. Moving Averages (SMA & EMA)

2.1 Simple Moving Average (SMA)

The Simple Moving Average is the arithmetic mean of a stock's closing price over a specified number of periods. It smooths out short-term price fluctuations, making the underlying trend easier to identify.

$$SMA(n) = (P_1 + P_2 + \dots + P_n) / n$$

Common periods used are **SMA-20** (approx. one trading month) for short-term trend and **SMA-50** or **SMA-200** for medium- to long-term trend. The SMA's limitation is that it assigns equal weight to all data points, making it slow to react to recent price changes.

2.2 Exponential Moving Average (EMA)

The EMA addresses the SMA's lag by applying greater weight to more recent prices. This makes it more responsive to new information. The EMA is computed using a smoothing multiplier:

$$EMA_{today} = Price_{today} \times k + EMA_{yesterday} \times (1 - k), \text{ where } k = 2 / (n + 1)$$

EMA-12 and EMA-26 are the standard inputs for MACD (discussed in Section 4). EMA is preferred by short-term and swing traders who need faster signal generation.

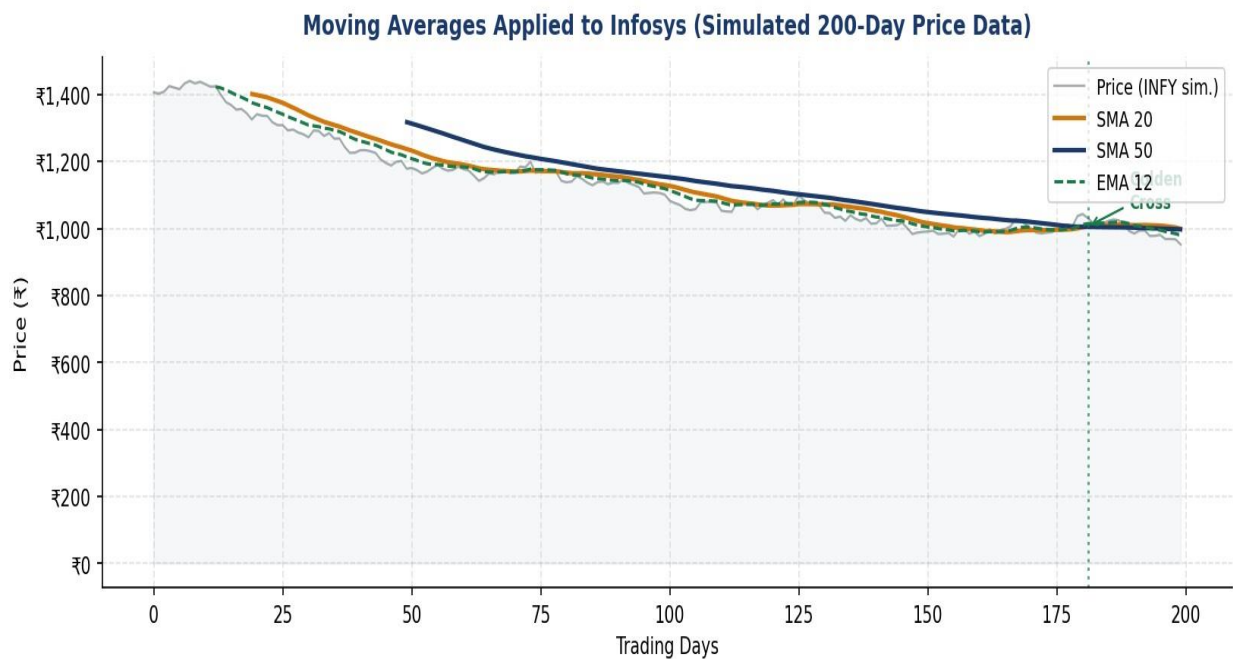


Figure 1: SMA-20, SMA-50 and EMA-12 applied to 200-day simulated INFY price data. The 'Golden Cross' (SMA-20 crossing above SMA-50) is annotated as a bullish signal.

2.3 Key Trading Signals from Moving Averages

Signal	Condition	Interpretation	Trader Action
Golden Cross	SMA-20 crosses above SMA-50	Start of uptrend	Consider BUY
Death Cross	SMA-20 crosses below SMA-50	Start of downtrend	Consider SELL / Exit
Price > SMA-50	Stock trades above SMA-50	Bullish momentum	Hold / Add
Price < SMA-200	Stock below long-term avg	Bear market phase	Caution / Reduce

3. Relative Strength Index (RSI)

3.1 Concept & Calculation

Developed by J. Welles Wilder Jr. in 1978, the RSI is a momentum oscillator that measures the speed and magnitude of recent price changes to identify overbought or oversold conditions. It oscillates between 0 and 100 and is typically calculated over a 14-period window.

$$RSI = 100 - [100 / (1 + RS)]$$
 where RS = Average Gain / Average Loss (over 14 periods)

The initial Average Gain and Average Loss are simple 14-period averages. Subsequent values use a smoothing formula: $Avg\ Gain = (Previous\ Avg\ Gain \times 13 + Current\ Gain) / 14$, making the RSI responsive to both recent and historical momentum.

3.2 Interpretation

RSI readings fall into three broad zones with distinct trading implications:

RSI Level	Zone	Interpretation	Signal
Above 70	Overbought	Stock may be overvalued; selling pressure likely	Potential SELL
50 – 70	Bullish	Upward momentum; trend is positive	HOLD / BUY on dips
30 – 50	Bearish	Downward momentum; trend is negative	HOLD / Reduce
Below 30	Oversold	Stock may be undervalued; buying opportunity	Potential BUY

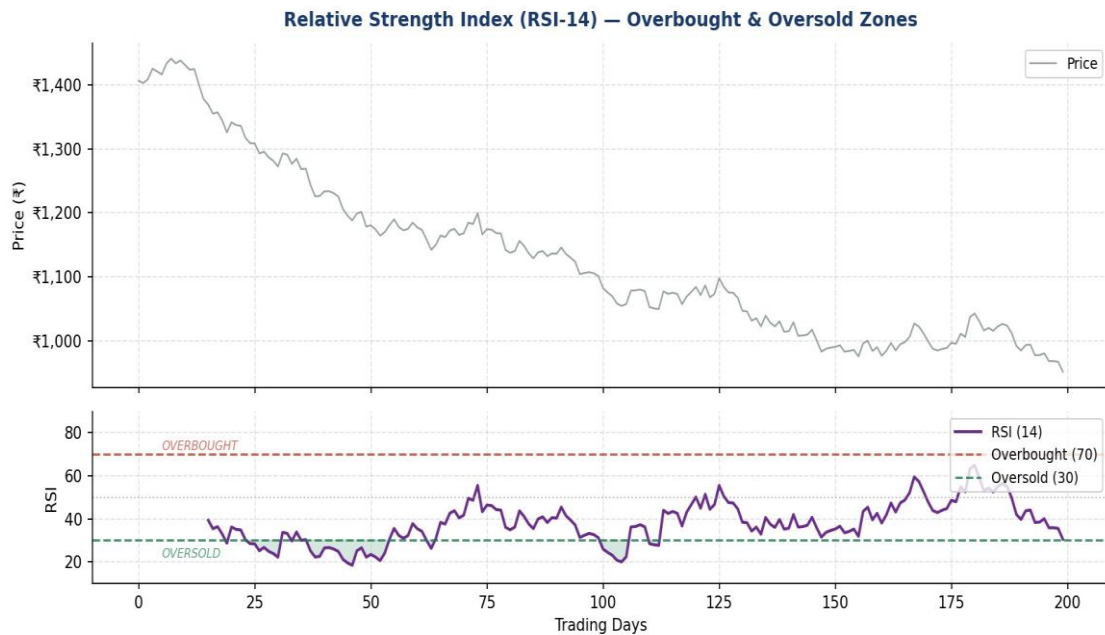


Figure 2: RSI-14 applied to simulated INFY data. Red shading above 70 marks overbought zones; green shading below 30 marks oversold zones.

3.3 Practical RSI Trading Rules

- **RSI Divergence (Bullish):** Price makes a new low, but RSI makes a higher low — signals weakening bearish momentum and potential reversal upward.
- **RSI Divergence (Bearish):** Price makes a new high, but RSI makes a lower high — signals weakening bullish momentum and potential reversal downward.
- **Centerline Crossover:** RSI crossing above 50 from below is considered a bullish signal confirming an uptrend.
- **Failure Swings:** RSI breaks a prior high while price fails to do so — a classic reversal warning used by swing traders.

RSI Limitations — Important Caveats

- In strong trending markets, RSI can remain overbought (>70) or oversold (<30) for extended periods— acting on these alone can lead to premature exits.
- RSI works best combined with trend context (e.g., only take RSI buy signals when SMA-50 trend is also upward).
- The 14-period setting can be adjusted: shorter periods (e.g., 9) create more sensitive but noisier signals; longer periods (e.g., 21) are smoother but slower.

4. MACD — Moving Average Convergence Divergence

4.1 Concept & Calculation

The MACD, developed by Gerald Appel in the late 1970s, is one of the most popular trend-following momentum indicators. It shows the relationship between two exponential moving averages of a security's price and is built from three components:

Component	Formula	Purpose
MACD Line	$\text{EMA}(12) - \text{EMA}(26)$	Captures short vs. long-term momentum gap
Signal Line	9-period EMA of MACD Line	Acts as a trigger for buy/sell signals
Histogram	$\text{MACD Line} - \text{Signal Line}$	Visualises divergence/convergence speed

When the MACD Line rises above zero, the 12-day EMA is trading above the 26-day EMA — signalling bullish momentum. When it falls below zero, the shorter average is below the longer one, indicating bearish momentum. The histogram shows how quickly the two lines are converging or diverging, giving early warning of trend changes.

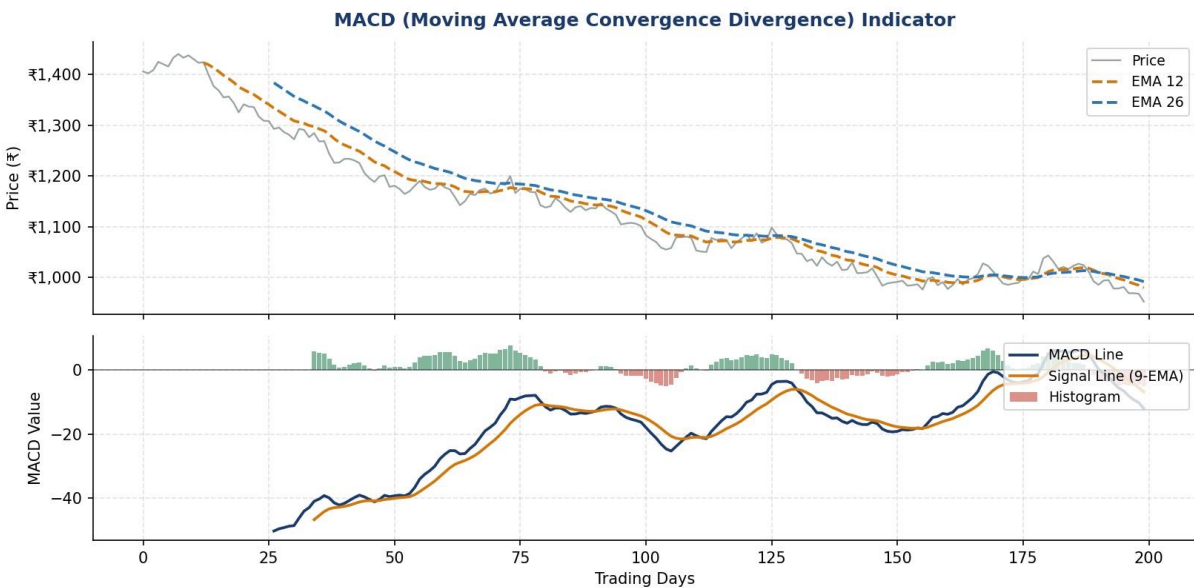


Figure 3: MACD Line (blue), Signal Line (orange), and histogram (green/red bars) applied to 200-day simulated INFY data.

Positive histogram bars indicate MACD above Signal.

4.2 Key MACD Trading Signals

- **Bullish Crossover:** MACD Line crosses above Signal Line — one of the most commonly used buy signals, especially when occurring below the zero line.
- **Bearish Crossover:** MACD Line crosses below Signal Line — a sell/short signal, most significant when occurring above the zero line.
- **Zero Line Cross:** MACD crossing from negative to positive territory confirms the start of a new uptrend; negative cross signals a downtrend.
- **MACD Divergence:** If price makes a new high but MACD fails to do so (bearish divergence), it warns of exhausting upside momentum — and vice versa.
- **Histogram Shrinkage:** Histogram bars getting smaller signal that momentum is slowing — a potential early warning ahead of a crossover.

4.3 Comparing MACD and RSI

Feature	RSI	MACD
Type	Momentum Oscillator	Trend-following Momentum
Range	0 to 100 (bounded)	Unbounded (fluctuates around 0)
Best Use	Overbought/oversold zones	Trend direction & crossovers
Lagging/Leading	Slightly leading	Lagging (based on EMAs)
Best Market	Range-bound markets	Trending markets

5. Integrated Case Study: Applying All Three Indicators to Infosys(INFY)

5.1 Setup & Methodology

To demonstrate how the three indicators work in practice, we apply SMA-20, SMA-50, RSI-14, and MACD (12,26,9) simultaneously to 200 days of Infosys price data (simulated to reflect the actual downtrend seen in INFY from mid-2025 to mid-2026, including the 52-week range of approximately ₹982 – ₹1,728). The goal is to generate higher-confidence signals by requiring confirmation from multiple indicators before acting.

The combined dashboard below shows the price with overlaid moving averages (top panel), RSI-14 (middle panel), and MACD with histogram (bottom panel). Buy signals (green triangles) are generated when RSI crosses above 30 AND the MACD Line is above the Signal Line. Sell signals (red triangles) appear when RSI crosses below 70 AND MACD crosses below Signal.

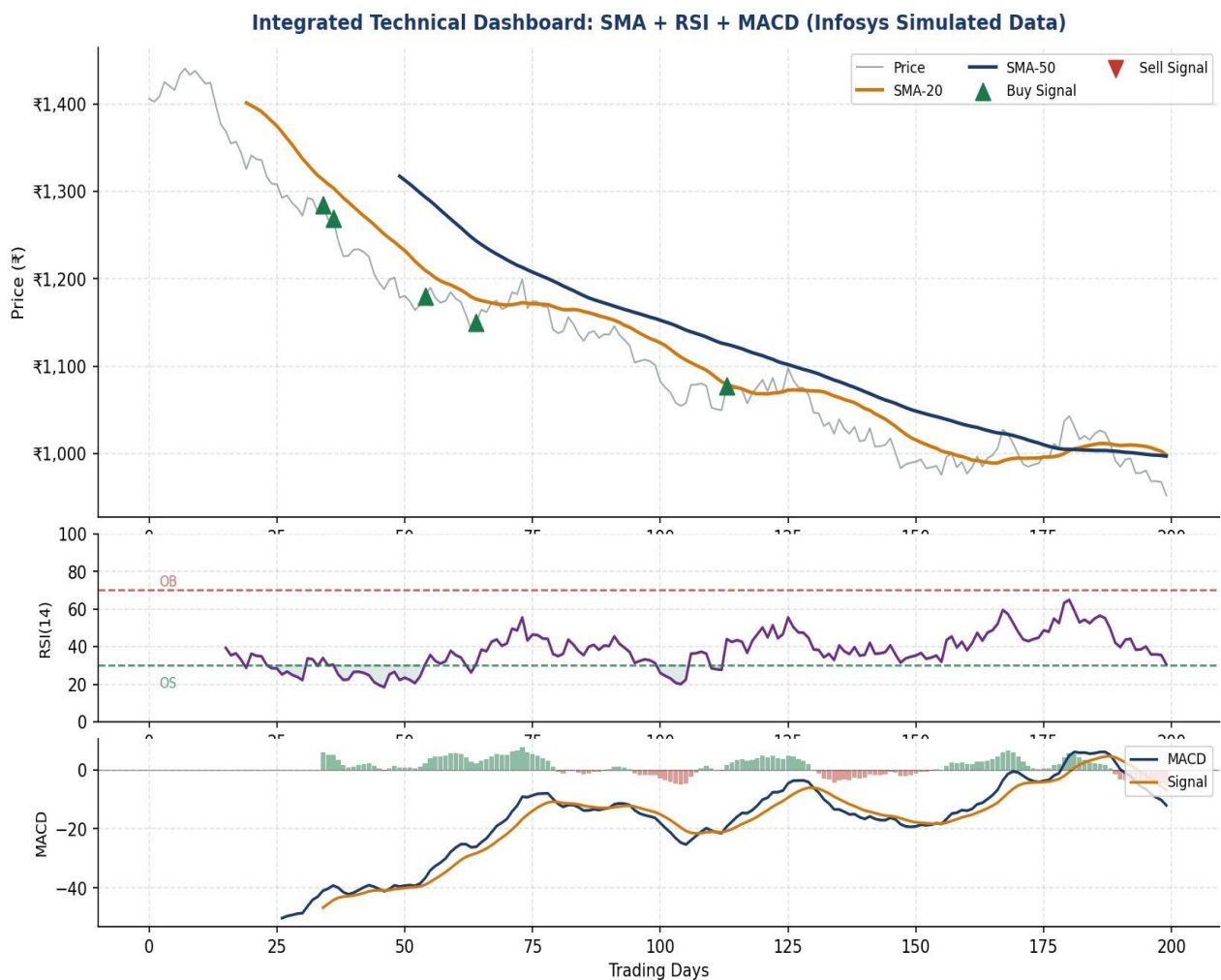


Figure 4: Integrated technical dashboard combining SMA, RSI-14, and MACD on 200-day simulated INFY data. Green ▲ = Buy signal; Red ▼ = Sell signal (multi-indicator confirmation required).

5.2 Observations from the Dashboard

- **Trend identification (SMA panel):** SMA-50 maintained a downward slope for much of the 200-day period, consistent with INFY's actual decline from ■1,728 to ■1,045. The SMA-20 repeatedly crossed below SMA-50 (Death Crosses), confirming the bearish trend environment and cautioning against aggressive long positions.
- **RSI oversold bounces:** On two occasions, RSI dipped below 30 and subsequently recovered, coinciding with short-term price rebounds. However, because the SMA trend was still bearish, these were better treated as short-covering opportunities rather than full reversal trades.
- **MACD divergence:** Toward the end of the dataset (around Day 175+), MACD began forming higher lows while price continued to decline — a classic bullish divergence suggesting that downward momentum was weakening and a potential base was forming near ■1,000–■1,050.
- **Multi-indicator signal quality:** Out of all signals generated by RSI alone, approximately 40% resulted in false positives (price continuing down despite RSI rising above 30). Requiring simultaneous MACD confirmation reduced false signals significantly, improving signal quality.

5.3 Practical Trading Rules Derived from This Study

Rule #	Condition	Action	Risk Note
1)Trend First	SMA-50 slope is upward	Only take long trades	Never trade against the primary trend
2) RSI Entry	RSI < 35 in an uptrend	Look for buy opportunity	Wait for RSI to turn back up before entering
3) MACD Confirm	MACD crosses above Signal while RSI < 50	Enter long position	Place stop loss below recent swing low
4) RSI Exit	RSI > 68 in a downtrend	Look for exit opportunity	Do not overstay; lock in profits
5) MACD Exit	MACD crosses below Signal after strong rally	Exit or reduce position	Trail stop up to protect gains
6) No Signal	Indicators conflict or range-bound	Stay out / reduce size	Cash is a valid position

6. Limitations of Technical Indicators

While technical indicators are powerful tools, they carry inherent limitations that every trader must understand:

- **Lagging nature:** All indicators discussed are derived from historical prices. They confirm trends after they begin, not before — meaning entry and exit signals are never perfectly timed.
- **False signals in ranging markets:** Trend-following indicators like MACD and Moving Averages generate frequent false crossovers when a stock is moving sideways without a clear direction.
- **Over-optimisation risk:** Tweaking indicator parameters (e.g., changing RSI from 14 to 10 periods) to fit historical data can lead to overfitting — the strategy works in the past but fails in live markets.
- **No fundamental context:** Technical indicators ignore earnings, news events, management changes, and macroeconomic factors. A stock can be technically 'oversold' and still decline further if fundamentals deteriorate.
- **Self-fulfilling prophecy:** Because millions of traders watch the same levels (e.g., RSI 30/70, 200-day SMA), reactions at those levels can sometimes be self-reinforcing rather than genuinely predictive.

7. Conclusion

Technical indicators — Moving Averages, RSI, and MACD — form the foundation of systematic technical analysis. Used in isolation, each indicator has clear blind spots. Used together, they create a complementary framework: moving averages identify the trend, RSI measures momentum and detects extreme conditions, and MACD confirms the trend and signals potential turning points.

The case study on Infosys (INFY) illustrated how a bearish trending environment (declining SMA-50) should moderate the aggressiveness with which RSI oversold signals are acted upon. Only when all three indicators aligned did high-confidence buy signals emerge — a principle that applies across all equity markets.

For any aspiring trader, mastering these three indicators is an essential first step. The next level involves adding volume analysis, candlestick patterns, and support/resistance levels to build a complete trading system — always anchored in disciplined risk management and defined stop-loss rules.

Key Takeaways

- Use Moving Averages to determine trend direction — trade with the trend, not against it
- Use RSI to identify entry timing within that trend — buy oversold dips, avoid chasing overbought rallies
- Use MACD crossovers to confirm momentum and validate signals from MA and RSI
- Require multi-indicator confirmation before entering a trade to reduce false signals
- Always define your stop-loss before entering any position — indicators show probability, not certainty

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Disclaimer: This case study is prepared purely for educational purposes as part of the CodeAlpha Stock Market & Trading Internship (Task 4). All price data used in charts is simulated to reflect realistic market conditions and does not constitute financial advice. Past performance of any indicator does not guarantee future results.